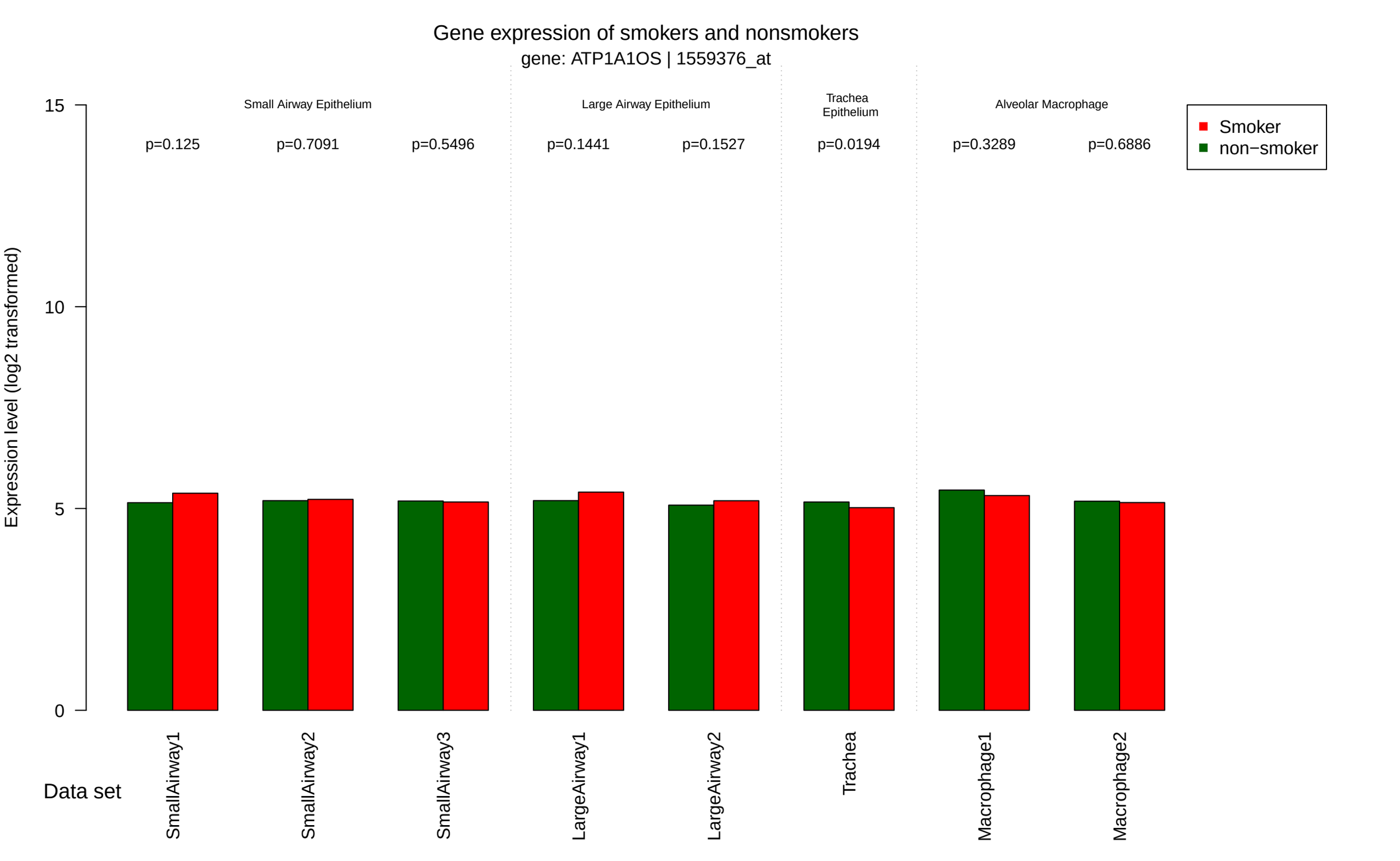
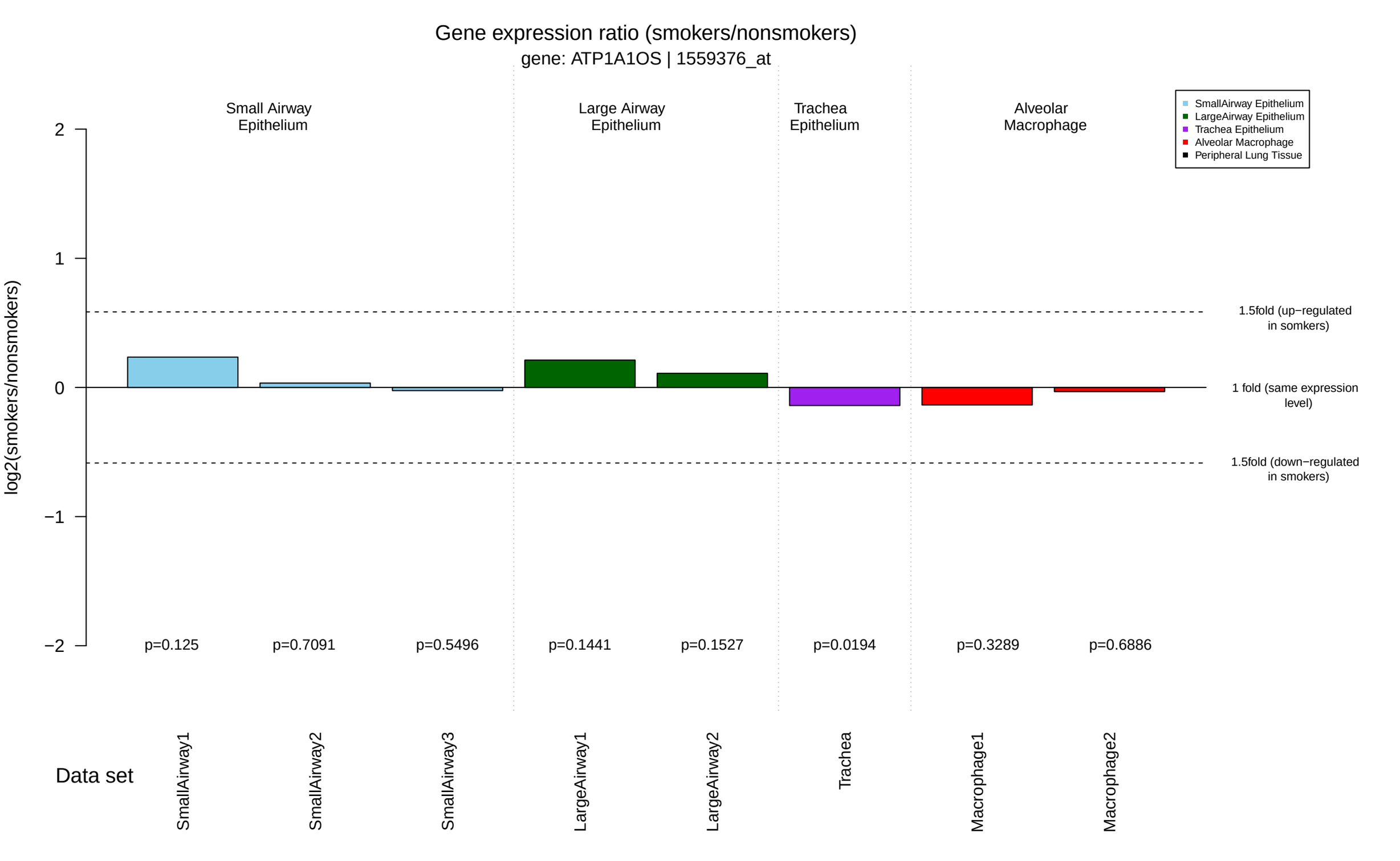
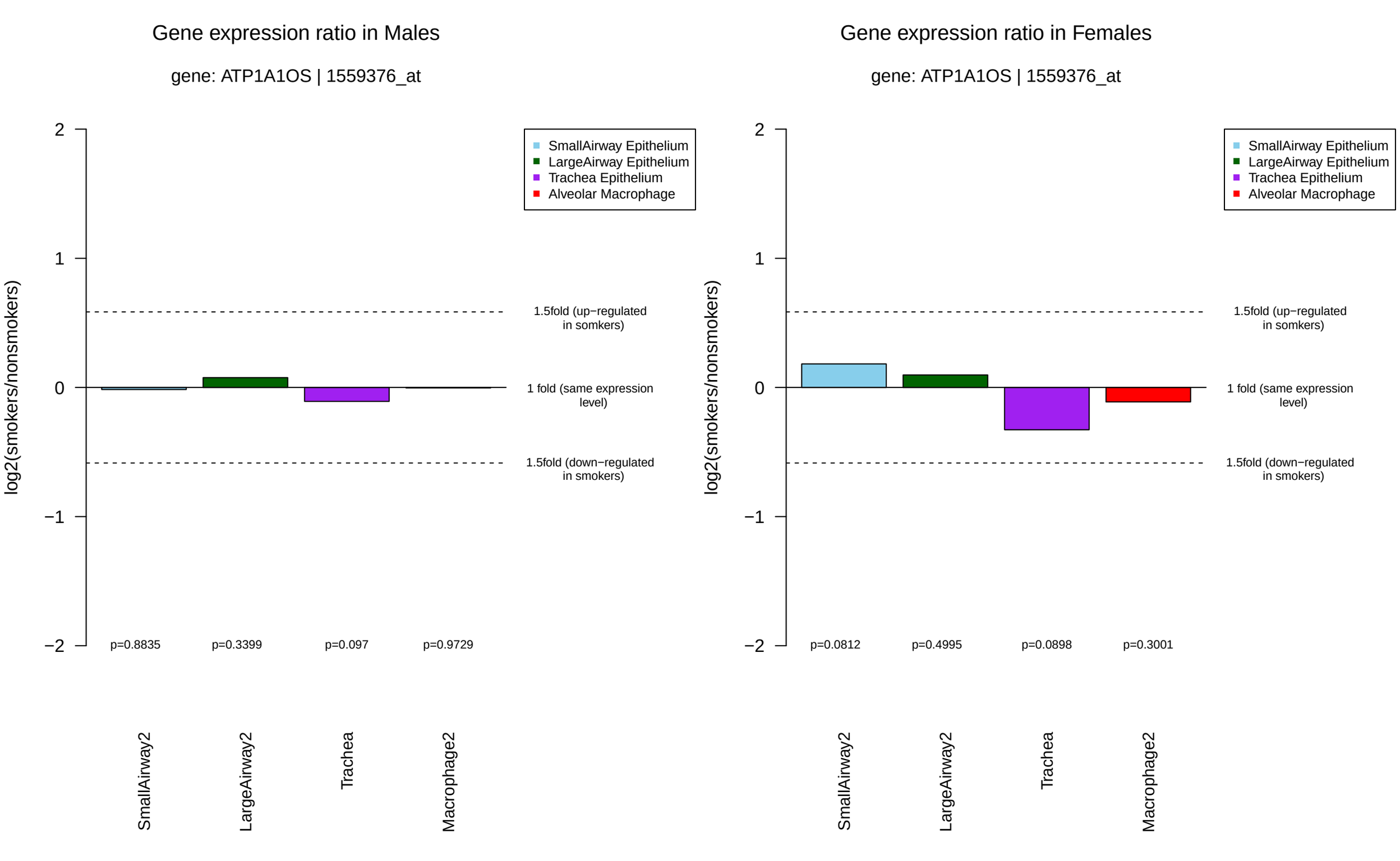
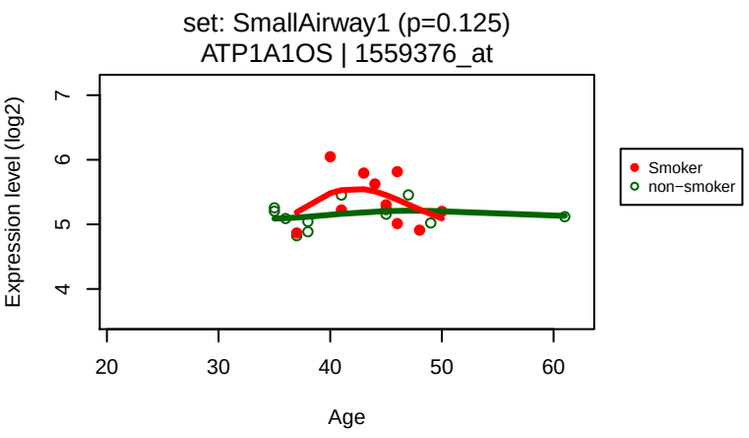
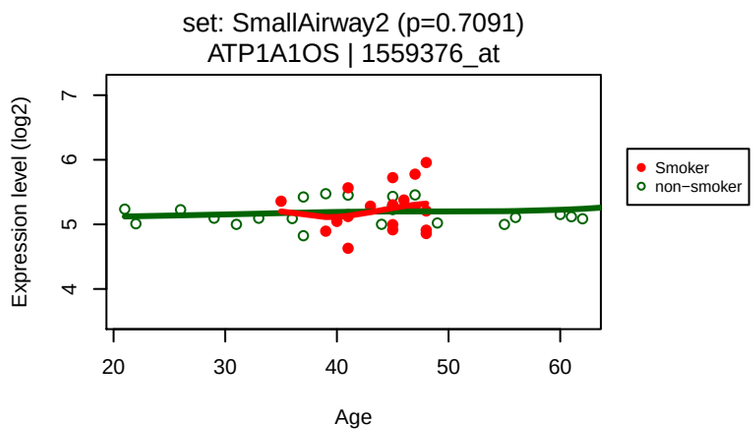
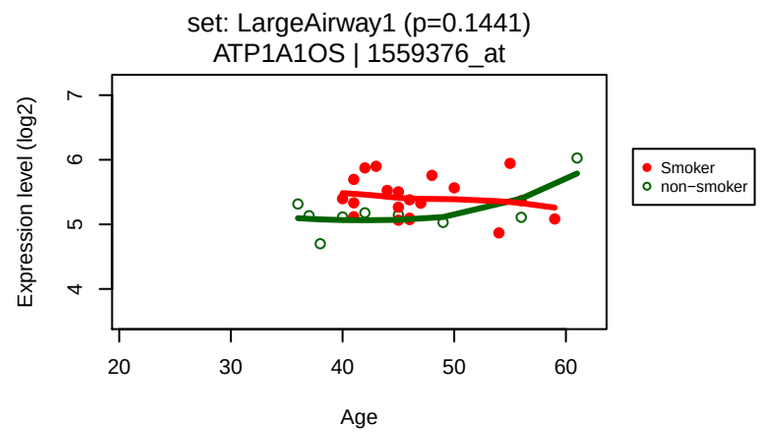

Gene expression of smokers and nonsmokers
gene: ATP1A1OS | 1559376_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.125
p=0.7091
p=0.5496
p=0.1441
p=0.1527
p=0.0194
p=0.3289
p=0.6886
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

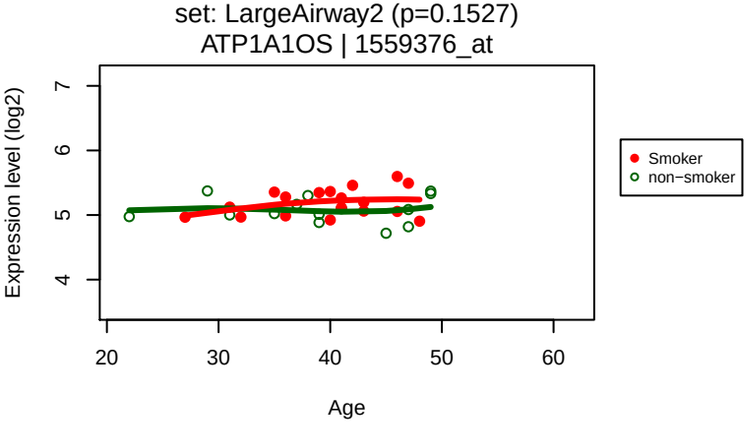

Gene expression ratio (smokers/nonsmokers)
gene: ATP1A1OS | 1559376_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.125
p=0.7091
p=0.5496
p=0.1441
p=0.1527
p=0.0194
p=0.3289
p=0.6886
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

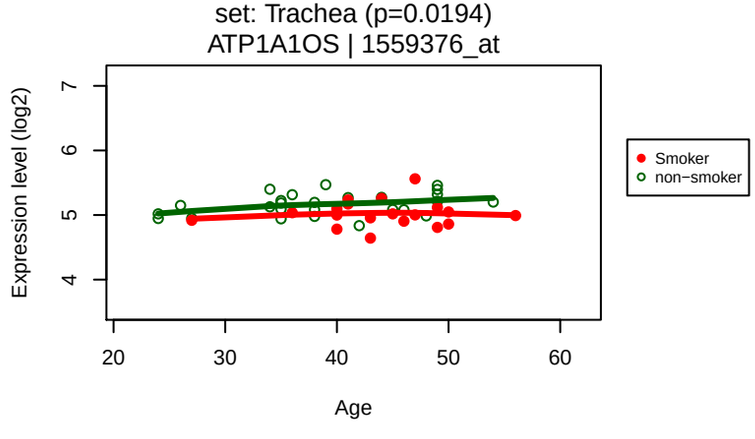

Gene expression ratio in Males
gene: ATP1A1OS | 1559376_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8835
p=0.3399
p=0.097
p=0.9729
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATP1A1OS | 1559376_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0812
p=0.4995
p=0.0898
p=0.3001
SmallAirway2
LargeAirway2
Trachea
Macrophage2

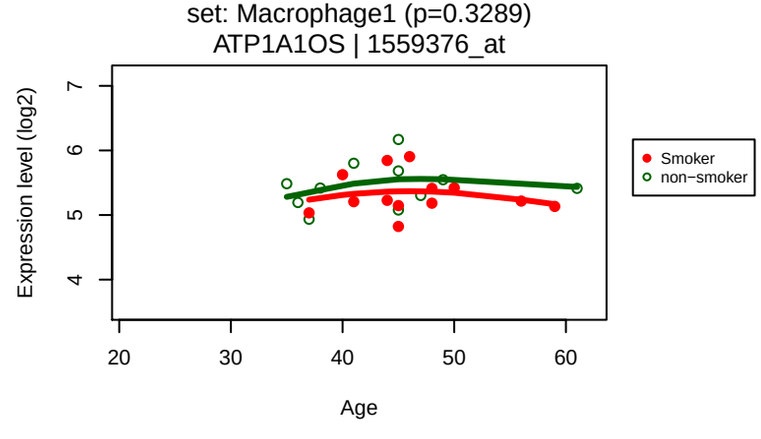

set: SmallAirway1 (p=0.125)
ATP1A1OS | 1559376_at
Expression level (log2)
Age
Smoker
non−smoker

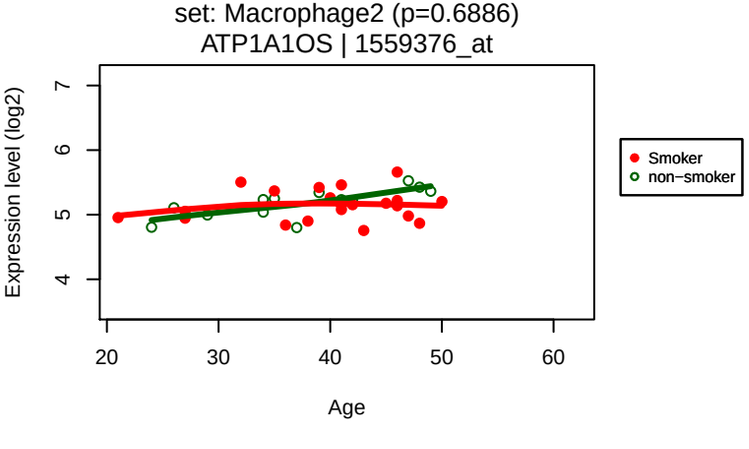

set: SmallAirway2 (p=0.7091)
ATP1A1OS | 1559376_at
Expression level (log2)
Age
Smoker
non−smoker

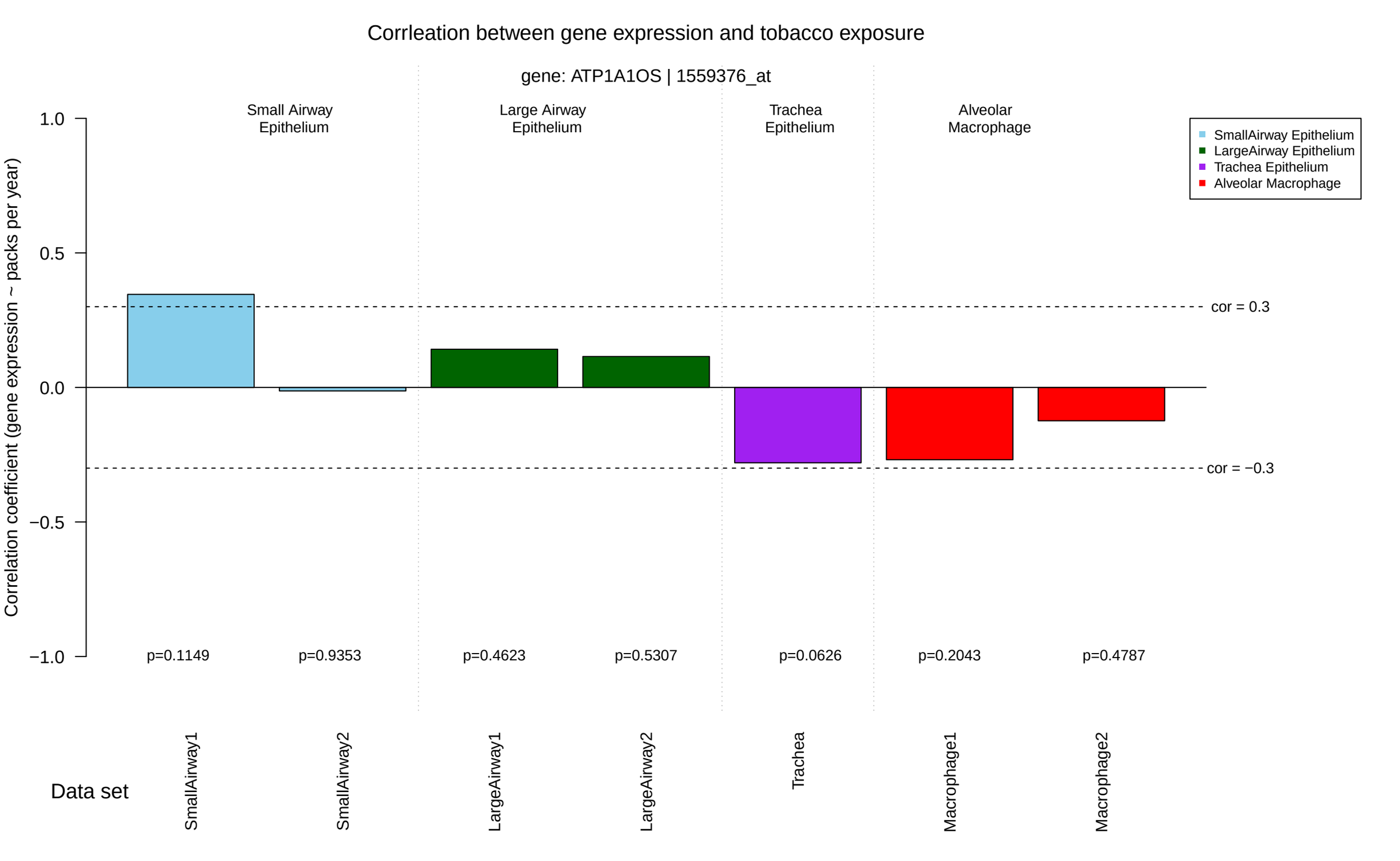

set: LargeAirway1 (p=0.1441)
ATP1A1OS | 1559376_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1527)
ATP1A1OS | 1559376_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0194)
ATP1A1OS | 1559376_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3289)
ATP1A1OS | 1559376_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6886)
ATP1A1OS | 1559376_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATP1A1OS | 1559376_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1149
p=0.9353
p=0.4623
p=0.5307
p=0.0626
p=0.2043
p=0.4787
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2